



KAIMOSI FRIENDS UNIVERSITY
Spring of Knowledge

PRELIMINARY TRAINING MANUAL

KAFU Digital Platform Administration & User Guide

System Administration Manual | Website User Manual

PART I
**System
Administration**

PART II
Website User Guide

ICT Directorate | Version 1.0 | June 2026 | CONFIDENTIAL

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System Administration Manual

For ICT Directorate staff responsible for deploying and maintaining the platform.

1. System Architecture Overview

The KAFU Digital Platform is a multi-component web application comprising four independently deployable units.

Component	Technology	Version	Deploy Path
Public Website	React + Vite + TypeScript	React 18, Vite 5	www.kafu.ac.ke
CMS Administration	React + Vite + TypeScript	React 18, Vite 5	cms.kafu.ac.ke
Staff Portal	React + Vite + TypeScript	React 18, Vite 5	portal-update.kafu.ac.ke
Backend API	Laravel + PHP	Laravel 11, PHP 8.2	api.kafu.ac.ke
Database	MySQL / MariaDB	MySQL 8.0+	Local to API server
Authentication	Laravel Sanctum	4.x	Token-based, per-app

NOTE

NOTE — All three React applications are Single Page Applications (SPAs) built to static files. They make API calls to the Laravel backend. The web server only serves static HTML/JS/CSS for the frontends.

2. Server & Hosting Requirements

Requirement	Specification
PHP	Version 8.2 or higher with extensions: mysql, mbstring, xml, curl, zip, gd, intl, fileinfo
Database	MySQL 8.0+ or MariaDB 10.6+
Web Server	Apache 2.4+ with mod_rewrite, mod_proxy, mod_headers enabled
Node.js	Version 18+ (for building frontend React applications)
pnpm	Version 8.x (monorepo package manager)

Requirement	Specification
Composer	Version 2.x (PHP package manager)
RAM	2 GB minimum — 4 GB recommended
Disk Space	10 GB minimum (for uploads, logs, and codebase)
SSL Certificate	Required for all subdomains — Let's Encrypt (free) or purchased certificate

3. Initial Server Setup

3a. Backend API Installation (Laravel)

- 1 Upload the API files**
Upload contents of artifacts/kafu-api/ to /var/www/kafu-api/ on the server. Exclude: .git/, node_modules/, database/database.sqlite
- 2 Install PHP dependencies**
Run: `composer install --no-dev --optimize-autoloader`
- 3 Configure environment**
Run: `cp .env.example .env` then edit .env with all production values (see Section 4).
- 4 Generate application key**
Run: `php artisan key:generate` — this encrypts sessions and tokens. Run ONCE per installation.
- 5 Set file permissions**
`chmod -R 755 /var/www/kafu-api && chmod -R 775 storage/ && chown -R www-data:www-data /var/www/kafu-api`
- 6 Run database migrations**
Run: `php artisan migrate --seed` — creates all tables and populates initial data.
- 7 Create storage symlink**
Run: `php artisan storage:link` — allows uploaded files to be publicly accessible.
- 8 Cache for production**
Run: `php artisan config:cache && php artisan route:cache && php artisan view:cache`

3b. Building & Deploying Frontend Applications

Run from the monorepo root directory:

1. Install all dependencies: `pnpm install`
2. Build Public Website (then upload dist/ to www.kafu.ac.ke):
`echo "VITE_API_URL=https://api.kafu.ac.ke" > artifacts/kafu-foundation/.env.production`
`pnpm --filter @workspace/kafu-foundation run build`
3. Build CMS Admin (then upload dist/ to cms.kafu.ac.ke):
`pnpm --filter @workspace/kafu-cms run build`
4. Build Staff Portal (then upload dist/ to portal-update.kafu.ac.ke):
`pnpm --filter @workspace/kafu-staff run build`

Apache .htaccess for each React SPA (required for client-side routing):

```
RewriteEngine On RewriteCond %{REQUEST_FILENAME} !-f RewriteCond %{REQUEST_FILENAME} !-d
RewriteRule ^ index.html [QSA,L]
```

4. Environment Configuration (.env)

IMPORTANT

IMPORTANT — Never commit the .env file to version control. It contains database passwords and the application encryption key.

Setting	Development Value	Production Value
APP_ENV	local	production
APP_DEBUG	true	false — CRITICAL for security
DB_CONNECTION	sqlite	mysql or mariadb
SESSION_SECURE_COOKIE	false	true — requires HTTPS
LOG_LEVEL	debug	warning
APP_URL	http://localhost	https://api.kafu.ac.ke

5. Database Administration

Command	Purpose	When to Use
php artisan migrate	Creates/updates all database tables	After initial setup or updates with new migrations
php artisan migrate --seed	Migrates + runs all seeders	Initial installation ONLY — will overwrite existing data
php artisan migrate:rollback	Reverses the last batch of migrations	To undo the most recent schema change
php artisan migrate:status	Shows which migrations have been run	Diagnostics
php artisan db:seed --class=X	Runs a specific seeder	Resetting specific configuration defaults

IMPORTANT

DANGER — NEVER run "php artisan migrate:fresh" or "migrate:fresh --seed" on the production database. These drop ALL existing data. Always take a backup first.

Database Backup (run regularly):

```
mysqldump -u kafu_db_user -p kafu_website > backup_$(date +%Y%m%d).sql
```

6. API Server Administration

Command	Purpose
<code>php artisan config:cache</code>	Cache config files — run after any <code>.env</code> change
<code>php artisan route:cache</code>	Cache route list — speeds up routing in production
<code>php artisan cache:clear</code>	Clear application cache
<code>php artisan queue:work</code>	Process background jobs (email notifications)
<code>php artisan schedule:run</code>	Run scheduled tasks (called by cron every minute)
<code>php artisan route:list</code>	Show all registered API routes with HTTP methods
<code>php artisan tinker</code>	Interactive PHP shell — admin use only

API Endpoint Reference:

Prefix	Purpose	Authentication
<code>GET /api/stats</code>	University statistics	Public
<code>GET /api/news, /api/events</code>	Published content listings	Public
<code>GET /api/navigation</code>	CMS-managed navigation menus	Public
<code>POST /api/cms/login</code>	CMS admin authentication	Public (rate-limited: 5/min)
<code>/api/admin/*</code>	CMS content management (full CRUD)	Sanctum token (CMS role)
<code>POST /api/staff/login</code>	Staff portal authentication	Public (rate-limited)
<code>/api/staff/*</code>	Staff profile management	Sanctum token (staff role)
<code>/api/reviewer/*</code>	Content review actions	Sanctum token (reviewer role)

7. CMS User Account Administration

Default Seeded Accounts — Change all passwords immediately before go-live:

Email	Default Password	Role
<code>super@kafu.ac.ke</code>	password	Super Admin
<code>ict@kafu.ac.ke</code>	password	ICT Admin

Email	Default Password	Role
comms@kafu.ac.ke	password	Communications Admin
reviewer@kafu.ac.ke	password	Reviewer
john.doe@kafu.ac.ke	password	Staff User (portal)

IMPORTANT

SECURITY — All seeded accounts use "password". Change EVERY account password before going live.

8. Scheduled Tasks & Cron Jobs

Only ONE cron entry is needed. Laravel handles all scheduling internally:

```
* * * * * php /var/www/kafu-api/artisan schedule:run >> /dev/null 2>&1
```

Scheduled Task	Frequency	Purpose
Publish Scheduled Content	Every minute	Auto-publishes items whose published_at datetime has passed
Expire Content	Hourly	Marks content as expired when expiry_date is passed
Prune Sanctum Tokens	Daily	Removes expired API authentication tokens
Database Backup	Weekly (Sun 02:00)	Auto-backup if configured

9. Monitoring, Logs & Troubleshooting

Log	Path	Contents
Laravel Application Log	storage/logs/laravel.log	PHP errors, exceptions, custom messages
Apache Access Log	/var/log/apache2/access.log	All HTTP requests to the API
Apache Error Log	/var/log/apache2/error.log	PHP fatal errors, config issues
CMS Audit Log	Database: audit_logs table	All CMS user actions (also in CMS Admin UI)

Common Issues & Solutions:

Symptom	Likely Cause	Solution
API returns 500 error	PHP exception or missing config	Check storage/logs/laravel.log. Set APP_DEBUG=true temporarily to see error details.
Website shows blank screen	React JS error or missing API	Open browser developer tools → Console. Check VITE_API_URL is correct.
API returns 403 Forbidden	File permissions or Apache config	chmod -R 755 /var/www/kafu-api && chmod -R 775 storage/. Check AllowOverride All.
Uploaded files not showing	Missing storage symlink	Run: php artisan storage:link
Changes not reflected	Cached config or routes	Run: php artisan config:clear && route:clear && cache:clear
Database connection refused	Wrong credentials in .env	Check DB_HOST, DB_USERNAME, DB_PASSWORD. Test: mysql -u kafu_db_user -p kafu_website

10. Security Hardening Checklist

- Change all default seeded passwords before going live
- Set APP_DEBUG=false in production
- Ensure APP_ENV=production
- Enable HTTPS on all subdomains (SSL certificate installed)
- Set SESSION_SECURE_COOKIE=true once HTTPS is live
- Configure CORS to only allow the three known frontend origins
- Set PHP upload limits: upload_max_filesize=20M, post_max_size=25M
- Disable PHP error display in production (display_errors = Off in php.ini)
- Keep Laravel, PHP, and all Composer packages up to date
- Block direct access to .env, .git/, and storage/ via Apache rules
- Use a dedicated MySQL user with only necessary database privileges (not root)
- Configure rate limiting on authentication endpoints (already set in Laravel)
- Enable Apache mod_headers and set security headers: X-Frame-Options, X-XSS-Protection, HSTS
- Review the CMS Audit Log monthly for unusual activity
- Take and test weekly database backups stored off-server

11. Maintenance Procedures

Applying System Updates:

- 1 Take a full backup**
Database dump + storage files. Never update without a restorable backup.
- 2 Enable maintenance mode**
In CMS → Site Controls → Maintenance Mode, or run: `php artisan down`
- 3 Pull the latest code**
`git pull origin main` in the project directory
- 4 Update PHP dependencies**
`composer install --no-dev --optimize-autoloader`
- 5 Run database migrations**
`php artisan migrate` (never `migrate:fresh` on production)
- 6 Rebuild frontends**
Rebuild and re-upload all three React apps
- 7 Clear and re-cache**
`php artisan config:cache && route:cache && cache:clear`
- 8 Disable maintenance mode & verify**
Test all three frontends and key API endpoints before announcing system is back.

12. Full Deployment Checklist

Infrastructure

- DNS records configured: A records for `www`, `api`, `cms`, `portal-update` all pointing to server IP
- SSL certificates installed and auto-renewing on all subdomains
- Apache virtualhost configured for each subdomain
- PHP 8.2+ installed with all required extensions
- MySQL/MariaDB running and accessible

Backend API

- Files uploaded (excluding `.git`, `node_modules`, `.env`)
- `composer install --no-dev` completed successfully
- `.env` configured with production values (not development defaults)
- `php artisan key:generate` run — `APP_KEY` is set
- Database created and credentials in `.env` are correct
- `php artisan migrate --seed` completed — all tables exist
- `php artisan storage:link` run — symlink created
- File permissions set: `storage/` and `bootstrap/cache/` are writable by `www-data`
- `php artisan config:cache && route:cache` run
- Cron job added for scheduler
- API health check: `https://api.kafu.ac.ke/api/stats` returns JSON (not HTML)

Frontend Applications

- Public website built with VITE_API_URL=https://api.kafu.ac.ke and uploaded
- CMS admin built and uploaded
- Staff portal built and uploaded
- .htaccess catch-all rules in place for all three SPAs
- www.kafu.ac.ke homepage loads correctly
- cms.kafu.ac.ke login page loads correctly
- portal-update.kafu.ac.ke login page loads correctly

Security & Go-Live

- All seeded default passwords changed
- APP_DEBUG=false confirmed
- HTTPS enforced — HTTP redirects to HTTPS on all subdomains
- CORS configured to allow only the three known origins
- Content reviewed by Communications — homepage and About page are accurate
- Smoke test: create news item in CMS → verify it appears on www.kafu.ac.ke/news
- Smoke test: staff login to portal → update a field → submit → verify in CMS review queue

PART II

Website User Manual

A complete guide to navigating www.kafu.ac.ke for students, staff, and the public.

13. Introduction to the KAFU Website

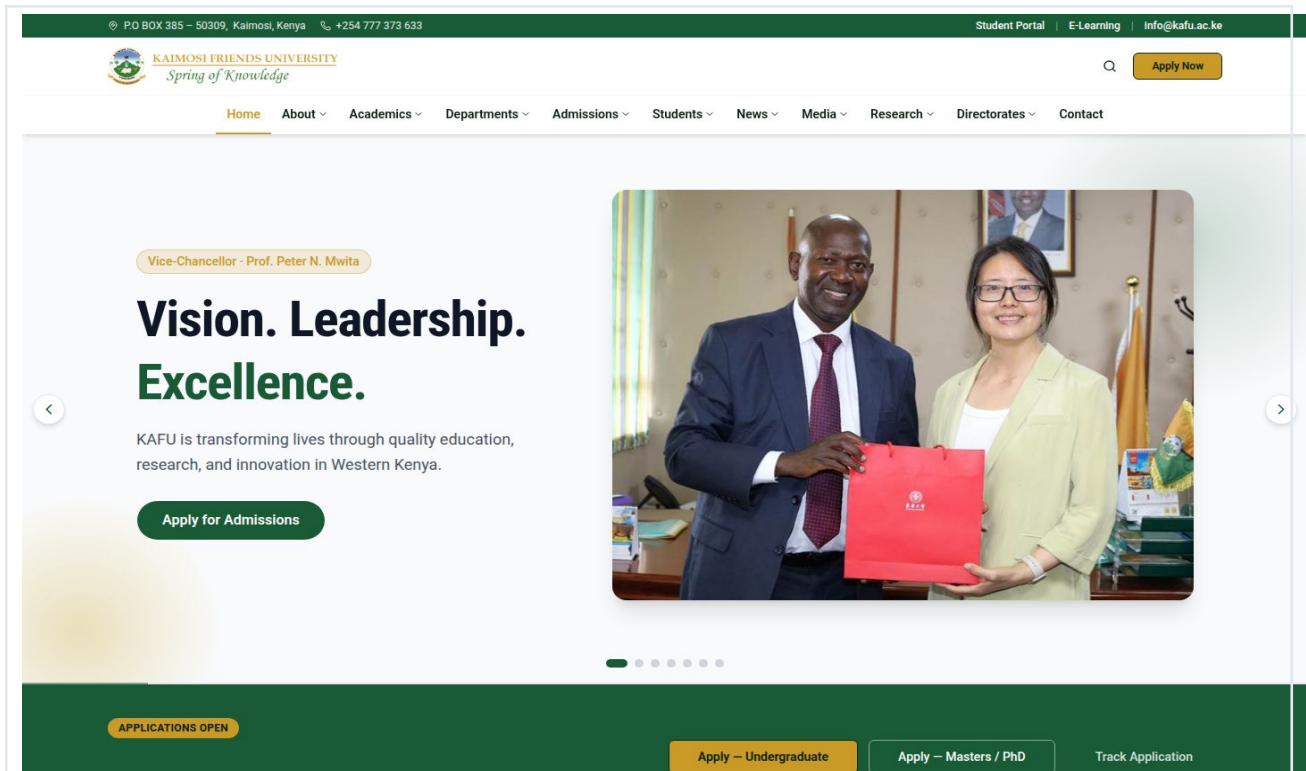


Figure 13.1 — The KAFU website homepage at www.kafu.ac.ke. The navigation bar provides access to all major sections.

The Kaimosi Friends University website is the official digital home of the university, providing comprehensive information about academic programmes, admissions, research, staff, news, and services.

14. Navigating the Website

Navigation Area	Contents
Utility Bar (top strip)	Contact info (phone, email) + Student Portal, E-Learning, Info@kafu.ac.ke quick links
Main Navigation	Logo (returns to homepage) + Home, About, Academics, Departments, Admissions, Students, News, Media, Research, Directorates, Contact

Navigation Area	Contents
Action Buttons	Search icon (magnifying glass) and gold "Apply Now" button (goes to admissions application)

Items with a dropdown arrow (▾) open mega-menu panels when clicked or hovered. The Search icon opens a full-text search across all content types.

15. Homepage

Section	What It Shows
Hero Slideshow	Rotating banner featuring key messages from university leadership
Admissions Banner	Current application status and countdown timers to key deadlines
Programme Discovery	Quick filter to find programmes by school and level
Schools & Faculties	Cards for each of the five schools linking to their detail pages
Latest News & Events	Four most recent published news articles and upcoming events
Opportunities	Current scholarships, job vacancies, tenders, and partnerships
Digital Services Hub	Quick links to Student Portal, E-Learning, Staff Portal, Library

16. About KAFU

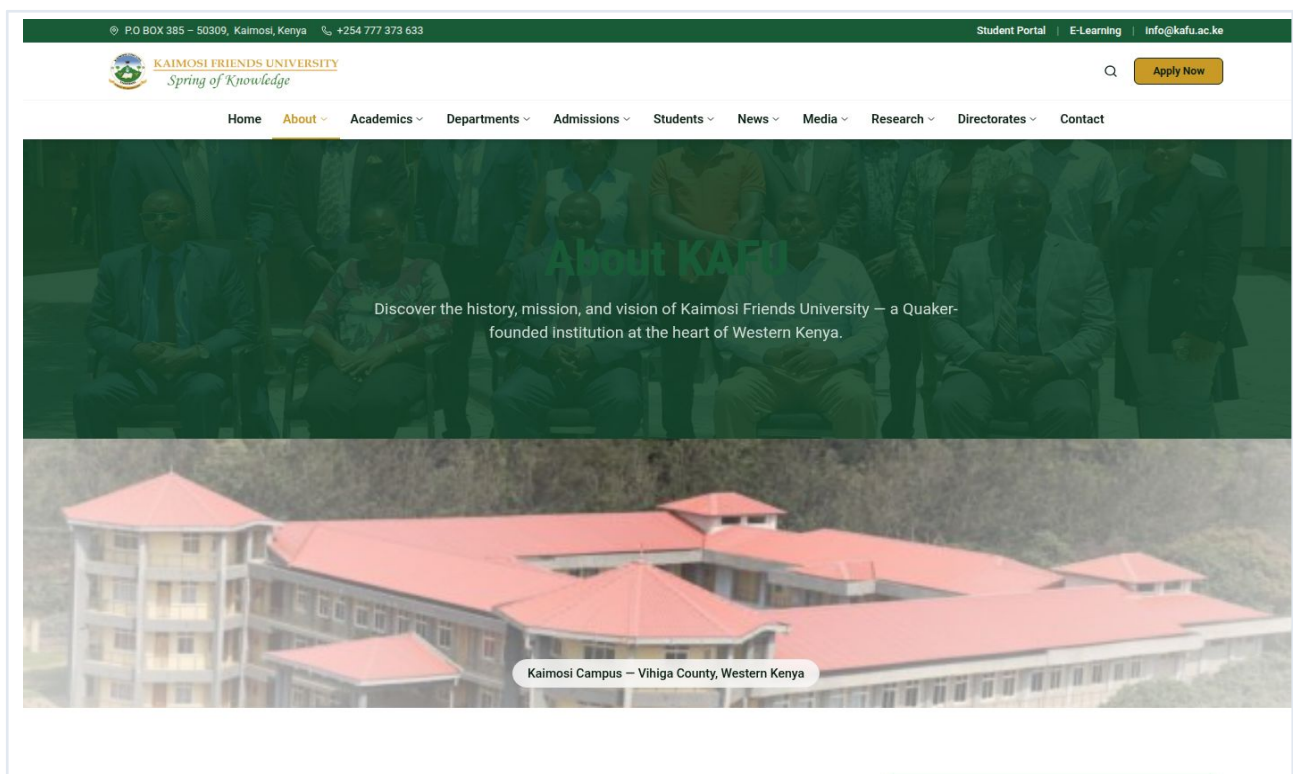


Figure 16.1 — The About KAFU page with campus imagery and institutional history.

About Sub-page	What to Find	URL
About KAFU	History, mission, vision, and core values	/about
Vice-Chancellor	Profile and message from the VC	/about/vice-chancellor
Management Board	University Management committee members	/about/management
University Council	Council members and governance structure	/about/council
Strategic Plan	Current strategic plan document	/about/strategic-plan
Policies & Regulations	University policies and regulations	/about/policies
Service Charter	Service delivery standards	/about/service-charter

17. Academics — Schools & Programmes

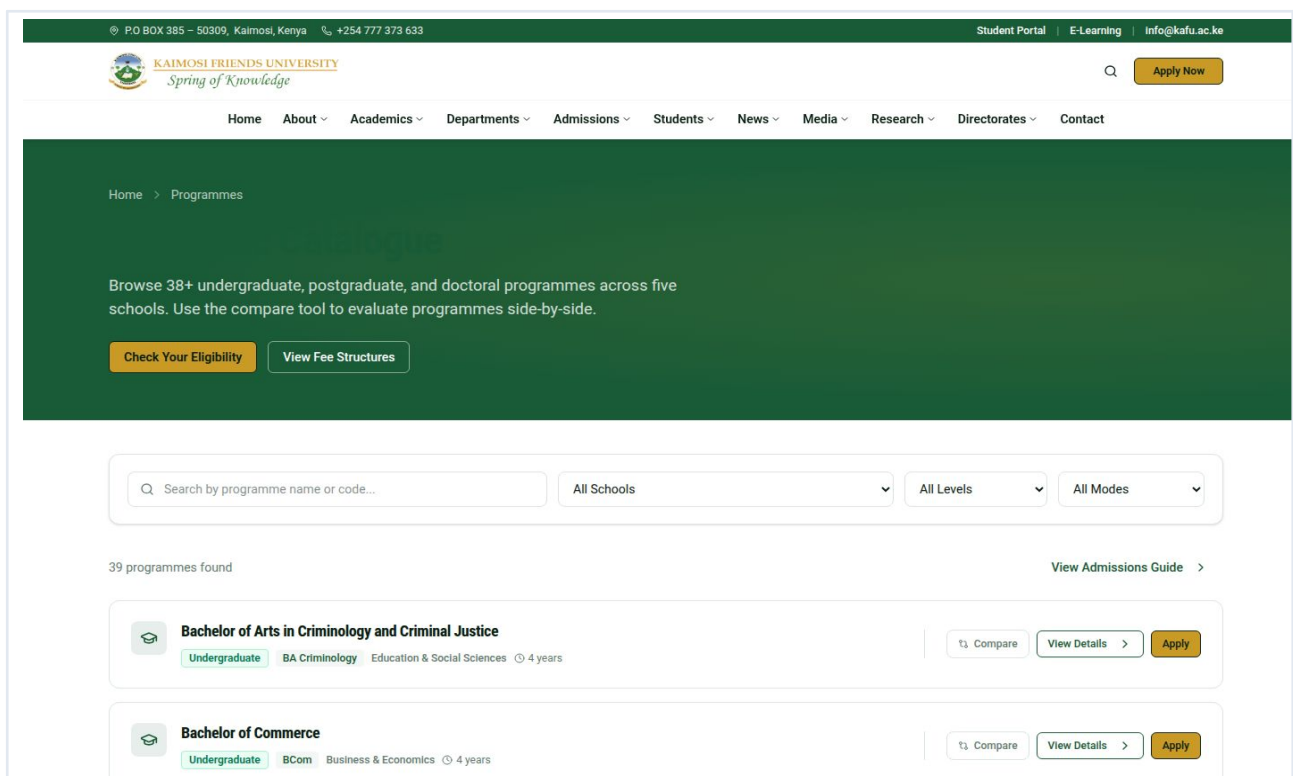


Figure 17.1 — The Programmes catalogue with 39 programmes, search, school and level filters, and a compare function.

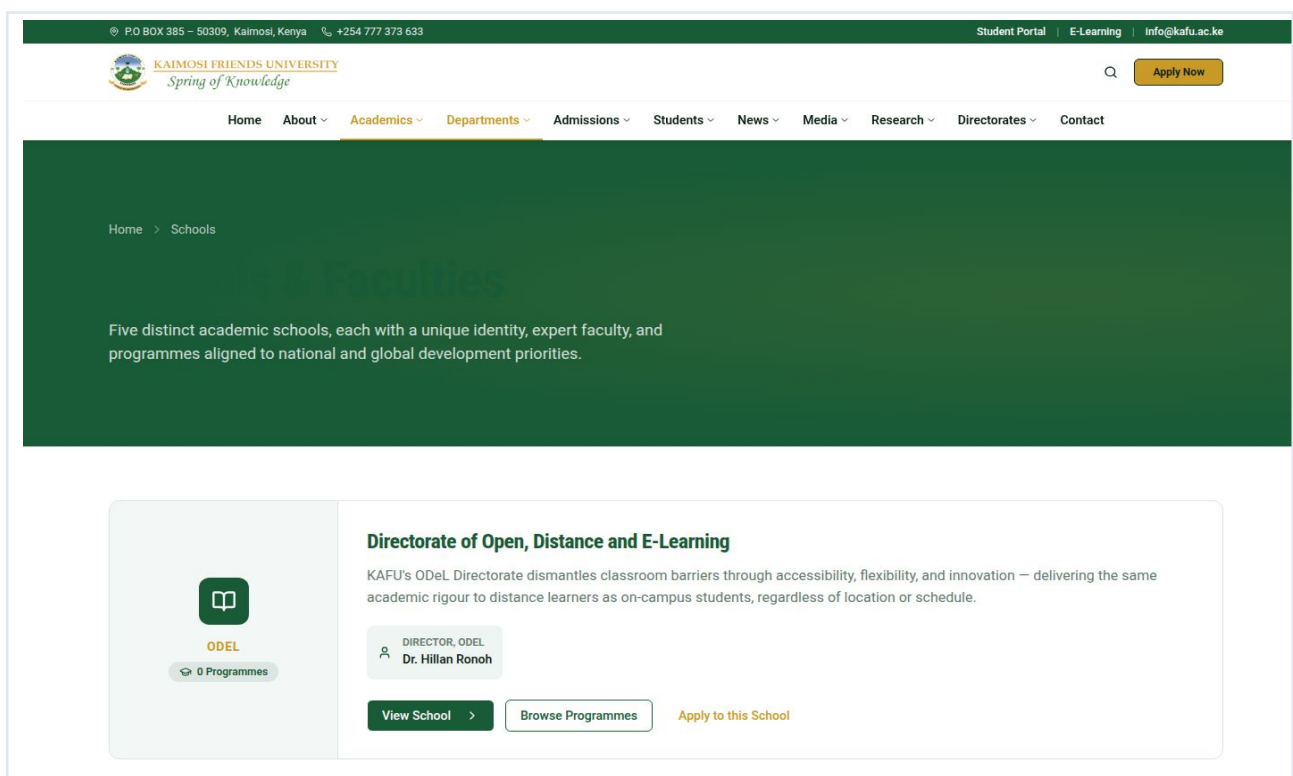


Figure 17.2 — The Schools page listing all five academic schools and the ODeL Directorate.

Code	School Name	Key Disciplines
SESS	School of Education & Social Sciences	Education, Curriculum Studies, Criminology, Psychology

Code	School Name	Key Disciplines
SBE	School of Business & Economics	Commerce, Business Administration, Finance, Economics
SCIT	School of Computing & Information Technology	Computer Science, Information Technology, Data Science
SOS	School of Science	Biology, Chemistry, Physics, Environmental Science
SHS	School of Health Sciences	Nursing, Public Health, Nutrition, Medical Sciences

Finding a Programme:

- 1 Go to Programmes**
Click Academics → All Programmes or go to /programmes.
- 2 Search or filter**
Type a keyword or use the School, Level, and Mode dropdowns.
- 3 View details**
Click "View Details" to see description, duration, entry requirements, and fees.
- 4 Compare**
Click "Compare" on up to 3 programmes then visit /programmes/compare to see them side by side.
- 5 Apply**
Click the gold "Apply" button to go to the online application portal.

18. Admissions

P.O BOX 385 – 50309, Kaimosi, Kenya
+254 777 373 633

Student Portal
E-Learning
info@kafu.ac.ke


KAIMOSI FRIENDS UNIVERSITY
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Home
About
Academics
Departments
Admissions
Students
News
Media
Research
Directorates
Contact

Home > Admissions

Join the Spring of Knowledge

Everything you need to apply to Kaimosi Friends University – from entry requirements and step-by-step guides to application forms and fee structures.

Apply Now
Verify KUCCPS Placement

KEY DATES & DEADLINES

CLOSED
KUCCPS Application Window

12 DAYS LEFT
Module II Applications Close

27 DAYS LEFT
Postgraduate Intake

31 JULY 2026
International Applications Close
31 July 2026

8 SEPTEMBER 2026
Reporting / Orientation
8 September 2026

Why Choose KAFU?

A Quaker-founded public university rooted in integrity, scholarship, and service to community.

Government Chartered

Strong Graduate Outcomes


Ready to Apply?
Applications for the 2025/2026

Figure 18.1 — The Admissions page with deadline countdown timers and application pathways.

Pathway	Who It Is For
KUCCPS (Government)	KCSE students placed through the Joint Admissions Board via KUCCPS
Module II (Self-Sponsored)	Self-sponsored undergraduate applicants not placed through KUCCPS
Postgraduate	First-degree holders applying for Masters or PhD programmes
International Students	Students from outside Kenya — see /international/study
KUCCPS Verification	Verify your KUCCPS placement status at KAFU — /kuccps-verify

TIP

TIP — Always check the Admissions page for the most current deadline dates — they change each academic year. Use the Eligibility Checker at /admissions/eligibility to check programme requirements.

19. News, Events & Announcements

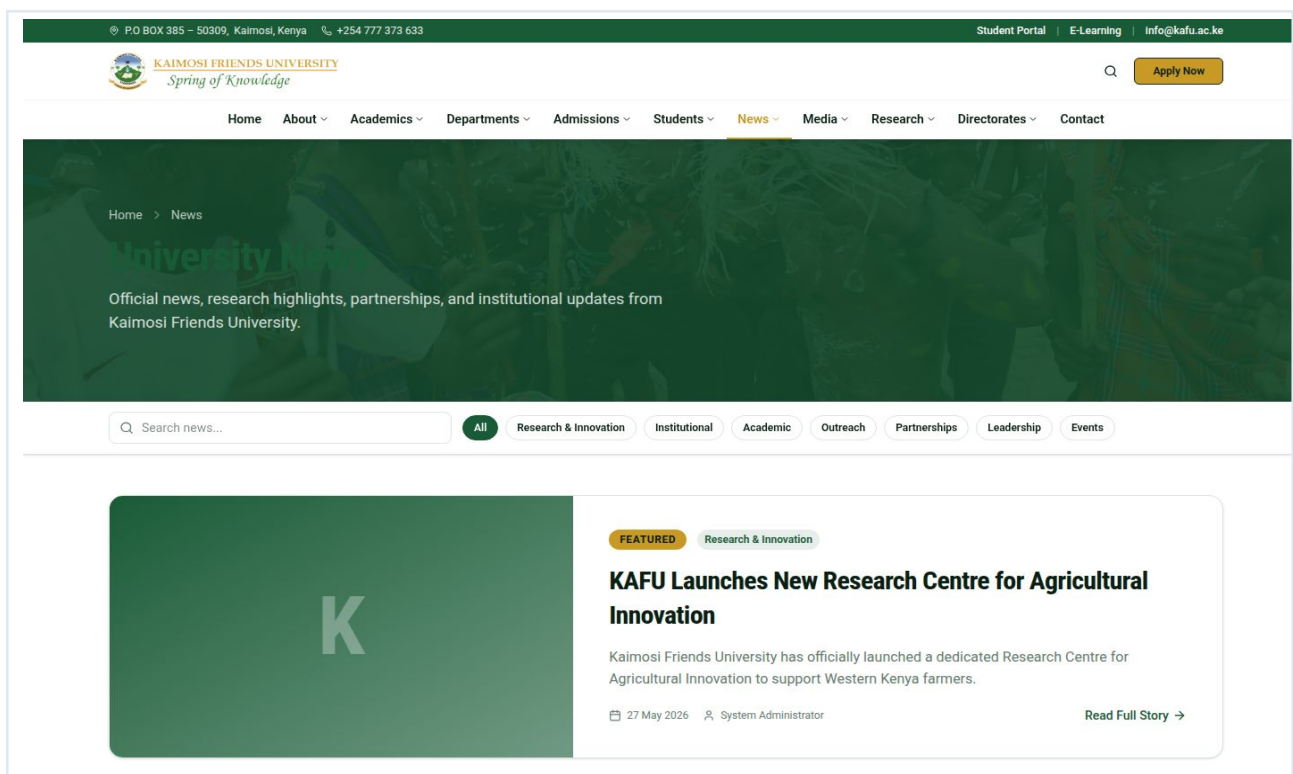


Figure 19.1 — The News page with category filter buttons. The top article is featured prominently.

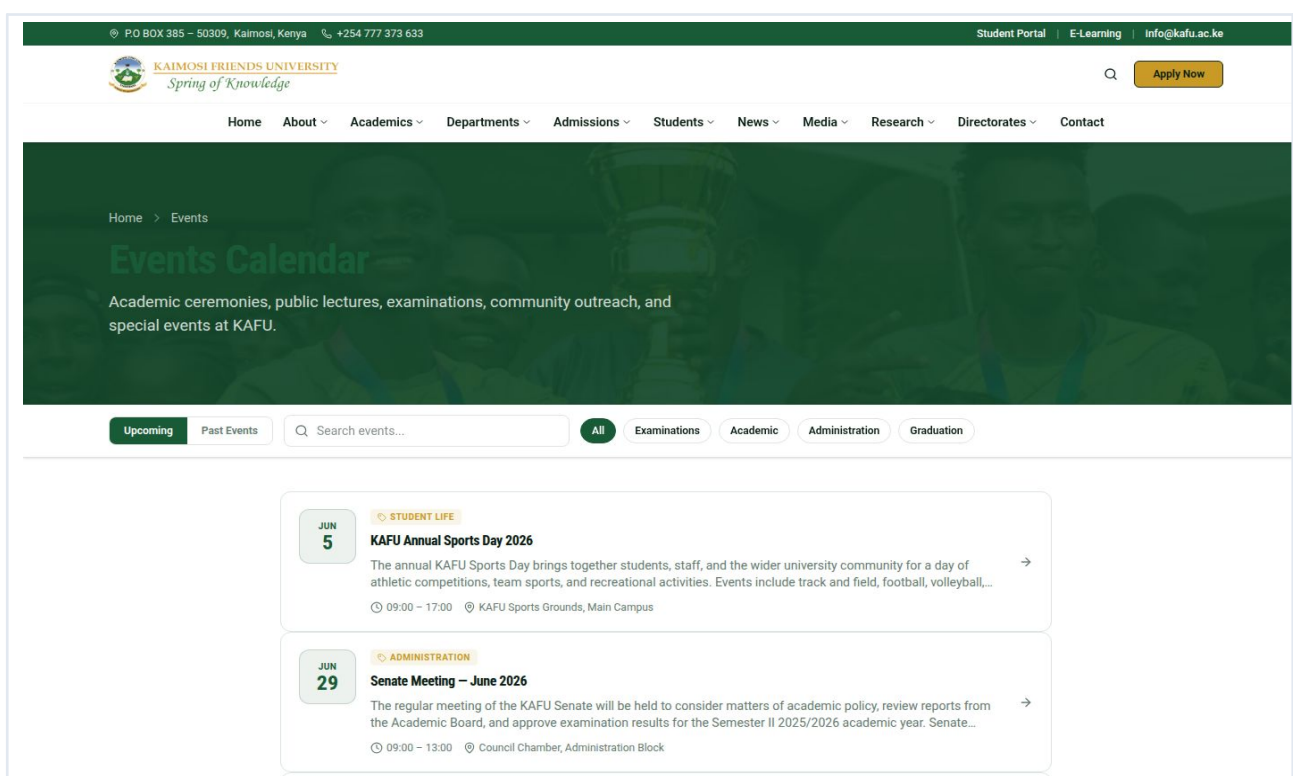


Figure 19.2 — The Events Calendar with Upcoming/Past toggle and event type filters.

News (/news): Use category filter buttons (Research, Institutional, Academic, etc.) and keyword search.

Events (/events): Toggle Upcoming/Past, filter by type, click any event for full details.

Announcements (/announcements): Official notices — exam timetables, fee notices, policy updates.

Opportunities (/opportunities): Scholarships, vacancies, tenders, and partnerships.

20. Research & Innovation

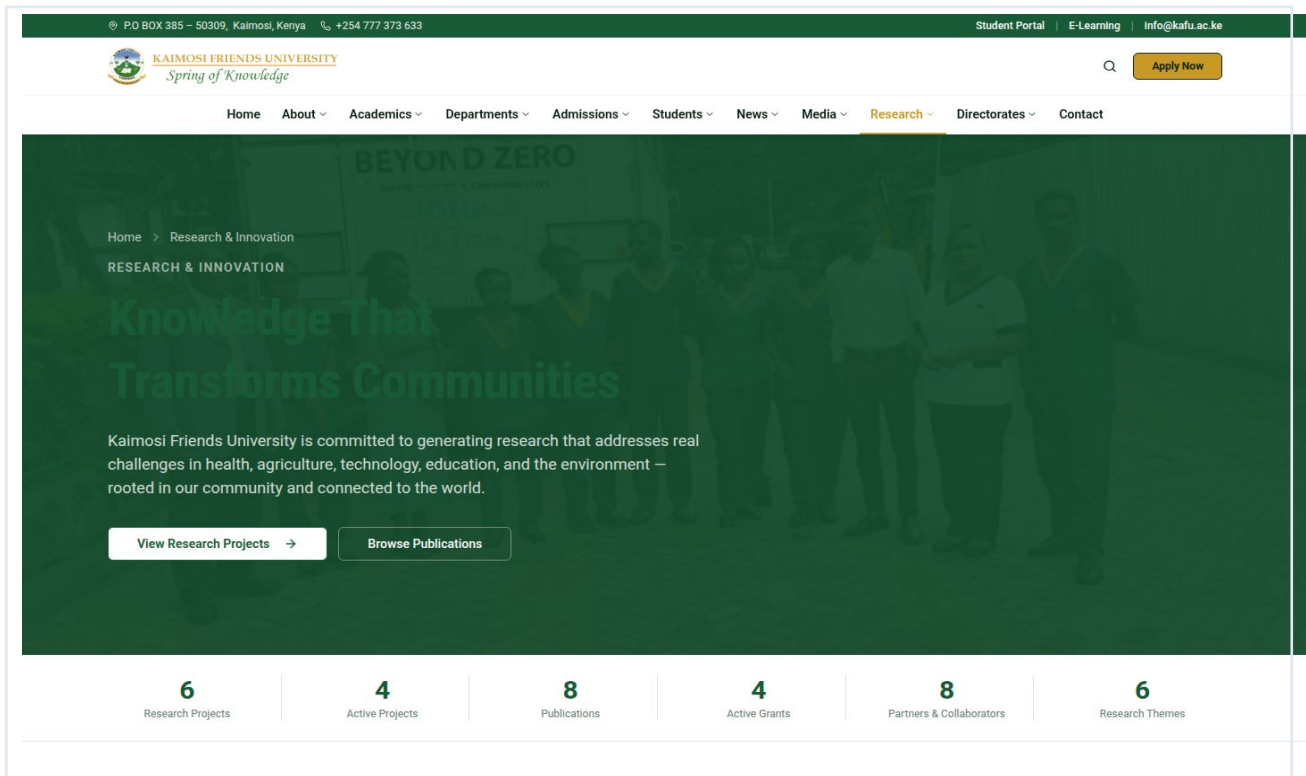


Figure 20.1 — The Research & Innovation page showing key statistics and focus areas.

Sub-section	What to Find	URL
Research Overview	Statistics, themes, and featured research achievements	/research
Research Projects	Active and completed projects with principal investigators	/research/projects
Publications	Searchable catalogue of papers, books, and conference proceedings	/research/publications
Partnerships	International and local research collaborations	/research/partnerships
KAFU Journal	University peer-reviewed academic journal	/research/journal

21. Staff Directory

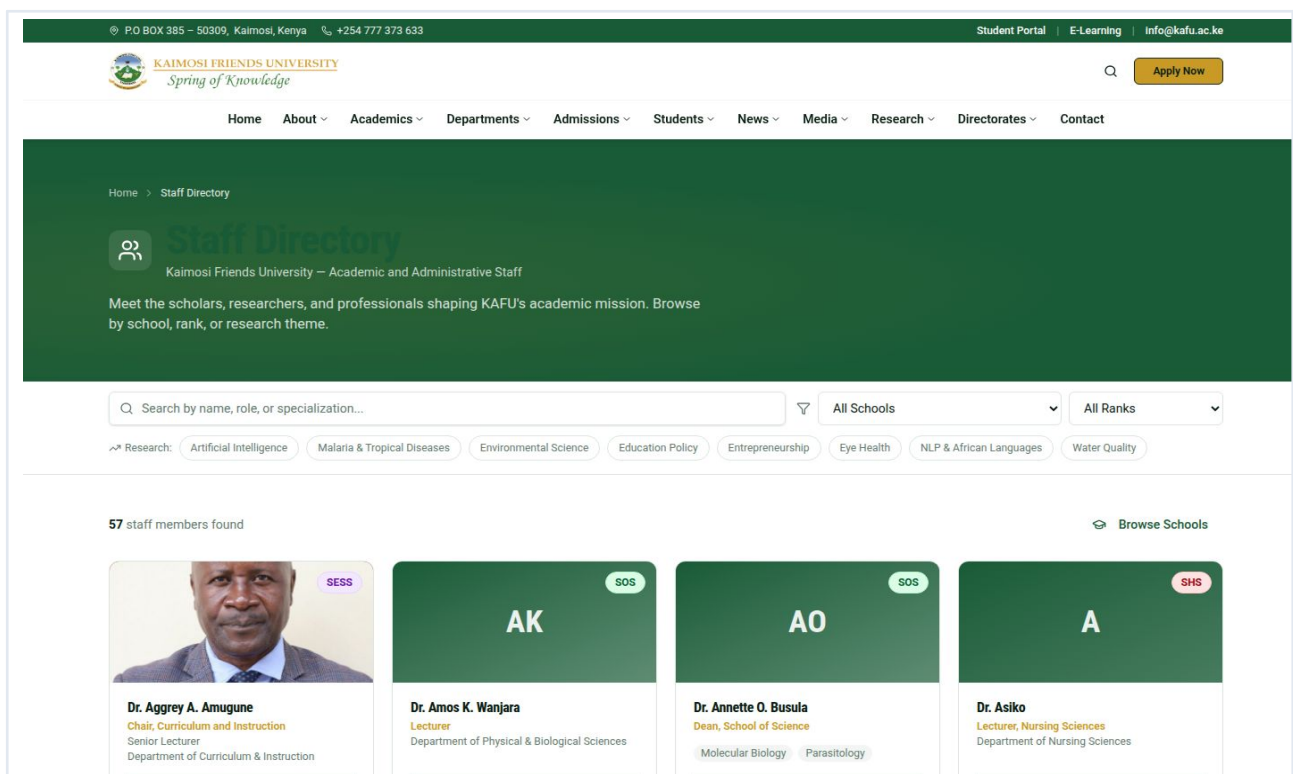


Figure 21.1 — The Staff Directory showing 57 academic staff with search, school filter, rank filter, and research theme tags.

- 1 **Use the search box**
Type a name, role (e.g. "Professor"), or specialisation (e.g. "Machine Learning").
- 2 **Filter by school**
Use the "All Schools" dropdown to see only staff in a specific school.
- 3 **Filter by rank**
Use the "All Ranks" dropdown for Professor, Senior Lecturer, Lecturer, etc.
- 4 **Browse research themes**
Click any tag (e.g. "Artificial Intelligence") to filter by research specialisation.
- 5 **Open a profile**
Click any staff card to see their full academic profile with bio, qualifications, publications, and contact details.

22. Contact & Campuses

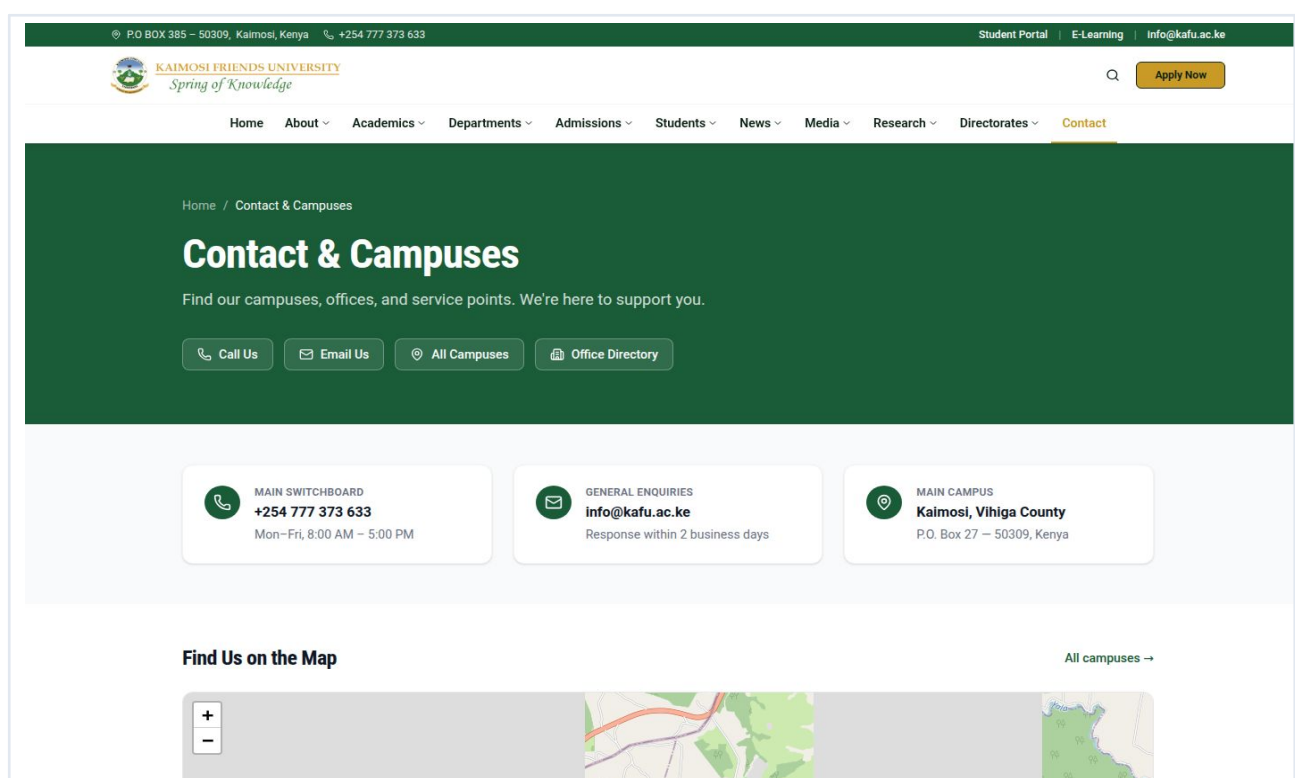


Figure 22.1 — The Contact & Campuses page with key numbers, email, location, and an interactive map.

Channel	Details	Hours
Main Switchboard	+254 777 373 633	Mon–Fri, 8:00 AM – 5:00 PM
General Enquiries	info@kafu.ac.ke	Response within 2 business days
Main Campus	Kaimosi, Vihiga County — P.O. Box 27–50309	—
ICT Support	ict@kafu.ac.ke	Mon–Fri, 8:00 AM – 5:00 PM
Admissions Office	admissions@kafu.ac.ke	Mon–Fri, 8:00 AM – 5:00 PM

23. Student Services & Digital Portals

Service	Purpose	How to Access
Student Portal	Academic records, registration, fee payments, exam results, timetables	portal.kafu.ac.ke — student number and password
E-Learning Platform	Course materials, assignments, online lectures, discussion forums	elearning.kafu.ac.ke — student email and password

Service	Purpose	How to Access
Library Catalogue	Search physical and digital library resources, reserve books	Students → Library or library.kafu.ac.ke
Student Services	Counselling, health services, clubs, accommodation, welfare	/student-services
International Students	Visa guidance, housing, orientation, support	/international

NOTE

IMPORTANT — The website at www.kafu.ac.ke is an information website only. To access your student records, register for units, pay fees, or download results, log in to the Student Portal at portal.kafu.ac.ke.

24. Accessibility & Browser Support

Browser	Minimum Version	Support
Google Chrome	Version 90+	Full support — recommended
Mozilla Firefox	Version 88+	Full support
Microsoft Edge	Version 90+	Full support
Safari (macOS/iOS)	Version 14+	Full support
Chrome (Android)	Latest	Full support — mobile responsive
Internet Explorer	Any version	Not supported

A. Glossary of Terms

Term	Definition
API	Application Programming Interface — the backend system the three React frontends call to fetch and update content.
Artisan	Laravel's command-line tool used by ICT to run migrations, clear caches, and manage the application from the server terminal.
CMS	Content Management System — the administration interface (cms.kafu.ac.ke) used by authorised staff to create and manage website content.
Composer	PHP package manager — used to install and update the Laravel API's dependencies.
CORS	Cross-Origin Resource Sharing — a browser security mechanism. Must be configured on the API to allow the React frontends to make requests.
Migration	A version-controlled database schema change. Running "php artisan migrate" applies all pending schema changes.
Monorepo	A single code repository containing all four applications. Managed with pnpm workspaces.
Sanctum	Laravel's API authentication system. Issues tokens to CMS and staff portal users when they log in.
SPA	Single Page Application — a web app where navigation happens in the browser without full page reloads.
KUCCPS	Kenya Universities and Colleges Central Placement Service — places students in universities after KCSE.
ORCID	Open Researcher and Contributor ID — a unique identifier for academic researchers.

B. Key Contacts Reference

Department	Email	Purpose
ICT Directorate	ict@kafu.ac.ke	System access, technical issues, CMS and staff portal accounts
Communications	communications@kafu.ac.ke	Website content corrections, news & media enquiries
Admissions Office	admissions@kafu.ac.ke	Application queries, entry requirements, deadlines

Department	Email	Purpose
Human Resources	hr@kafu.ac.ke	Staff profile corrections, employment verification
Research Office	research@kafu.ac.ke	Research publications, grants, partnerships
International Office	international@kafu.ac.ke	International student enquiries, exchange programmes
General Enquiries	info@kafu.ac.ke	General questions not covered above

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